

How AI Learns and Solves Problems

SMaiLE Project

Key Information

Target Group: 4 - 7 y.o.

Duration: 2.5 – 3 hours (Modular)

Key Learning Goals:

1. **Understanding AI:** See AI as a helpful tool, not magic.
2. **Human Role:** Understand that robots need people to learn.
3. **Curiosity:** Explore real-life examples (vacuums, smart speakers).
4. **Creativity:** Imagine and design a helpful robot.

Learning Outcomes

Students will be able to:

KNOWLEDGE & UNDERSTANDING:

- Understand what a robot or AI assistant is.
- Learn that AI can help with tasks but needs human guidance.
- Explore real-life examples of AI helpers.

SKILLS & ABILITIES:

- Storytelling and discussion about technology.
- Role-playing to understand instructions.
- Creative drawing to express ideas.

ATTITUDES & VALUES:

- Build curiosity about technology.
- Reflect on the difference between humans and machines.
- Develop responsibility in using tools.



European Dimension / Erasmus+ Connection

- **Early Digital Literacy:** Provides an accessible entry point to AI education.
- **Critical Thinking:** Encourages children to question technology's role.
- **Inclusion:** Uses familiar examples to include all children in the conversation.
- **Future Readiness:** Prepares young learners for a digital world.



1. Resources and Tools

- **Props:** Puppet or toy robot.
- **Materials:** Printable cards ("Robot guesses right/wrong"), Drawing paper, Pencils.
- **Media:** Short video of robots in everyday life (e.g., SciShow Kids).
- **Assessment:** Checklist (included).

Useful Links

- [Short video about robot helpers](#)

Classroom Support Materials

- Use a simple assessment rubric, a puppet or robot prop, and Padlet or another board for collecting children's final reflections.

Activity Overview

Phase	Time	Activity
Intro	20 min	Story: "The Robot Who Got It Wrong". Discussing robot mistakes.
Research	30 min	Real AI Helpers: Video and Voting Game ("Can a robot do this?").
Creative	40 min	Game & Drawing: "Robot Helper or Human Job?" and "Design Your Robot".
Reflection	20 min	Circle Time: Sharing drawings and Memory Box activity.

2. Introduction and Motivation

Activity 2.1: Story: The Robot Who Got It Wrong

Goal: Spark curiosity and show that robots need help.

- **Activity:** Teacher reads a story about a robot who gets confused (e.g., putting shoes in the fridge).
- **Discussion:** "Why did the robot get it wrong?" "How do people know better?"
- **Key Concept:** Robots are helpful but don't think like people.

3. Research and Learning

Activity 3.1: Exploring Real AI Helpers

Goal: Connect concepts to real life.

- **Video:** Watch a short clip about robots (vacuums, delivery bots).
- **Discussion:** "Can a robot vacuum find your lost socks?" (No). "Can a smart speaker give you a hug?" (No).
- **Voting Game:** Teacher shows cards (e.g., Hospital Robot). Children vote if it needs a person to help.



4. Creative Application

Activity 4.1: Robot Helper or Human Job?

Sorting Game:

- Teacher calls out tasks: "Tidy room", "Give a hug", "Paint a picture".
- Children move to zones: **Robot**, **Person**, or **Both**.

Activity 4.2: Drawing Task: Design Your Robot

Assignment: Invent your own robot helper.

- What does it do well? (e.g., carry heavy boxes).
- What does it need help with? (e.g., knowing where to go).

5. Reflection and Evaluation

Activity 5.1: Circle Time & Memory Box

- Children share their robot drawings.
- **Memory Box:** Add drawings/notes of "How AI helped today" (e.g., Mom used a map app).

Assessment Checklist

- Recognised robot roles vs. human roles.
- Described how their robot helps.
- Engaged with drawing/memory box.
- Shared ideas confidently.